

Written Report on Comparative Study of Transformer Architectures (BERT, GPT-2, T5)

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Introduction:

Transformers have revolutionized natural language processing, powering tasks such as summarization, translation, and question answering through their ability to model long-range dependencies. Among them, three major architectures dominate: encoder-only models like BERT, which excel at contextual understanding; decoder-only models like GPT, optimized for fluent text generation; and encoder-decoder models like T5, which balance comprehension with generation. In this report, we compare these architectures on the SQuAD v1.1 dataset, a benchmark for question answering, by fine-tuning BERT, GPT-2, and T5 on the same task. The goal is to analyze how their architectural differences affect performance, efficiency, and answer quality, while reflecting on their strengths, weaknesses, and applicability to real-world QA systems.

Part 1: Dataset & Model Training

Dataset

- **SQuAD v1.1** selected for **question answering**.
- Chosen because:
 - Widely used benchmark for QA.
 - Supports both **extractive** (span) and **generative** answers.
 - Allows fair comparison of encoder-only, decoder-only, and encoder-decoder models.

Preprocessing

- **Tokenization:** Hugging Face tokenizers (bert-base-uncased, gpt2, t5-small).
- **Formatting:**
 - GPT-2: *context + question* → *answer* (causal LM).
 - BERT: context+question pair → predict start/end span.
 - T5: "question: ... context: ..." → answer.
- **Splits:** 1000 training examples, 200 validation examples (subset for speed).
- **Max lengths:** 384 (BERT), 512 (T5/GPT-2).
- **Other preprocessing:** truncation, padding, aligning labels with input IDs.

Models & Training Setup

- **Decoder-only:** GPT-2 small.
- **Encoder-only:** BERT-base (for extractive QA).
- **Encoder-decoder:** T5-small.

Training settings:

- Epochs: 2 (demo runs).
- Optimizer: AdamW ($\text{lr} \approx 3\text{e-}5$).
- Batch size: 2–8 (depending on model, with gradient accumulation).
- Mixed precision (fp16) enabled.
- Hardware: GPU recommended.

Difficulties faced:

- GPU memory → solved with gradient accumulation & FP16.
- GPT-2 training required careful label masking.
- Limited dataset subset → models underfit slightly.

Part 2: Evaluation & Analysis

Metrics used

- **Exact Match (EM)** and **F1** (for extractive QA).
- **BLEU** (for generative answers).
- **Perplexity** (for GPT-2 only).

Illustrative Sample

Context: “The Eiffel Tower received 6.9 million visitors in 2016.”

Question: “How many visitors did the Eiffel Tower receive in 2016?”

Gold Answer: “6.9 million”

- **BERT:** → “6.9 million” (exact span).
- **T5:** → “6.9 million” (correct generative).
- **GPT-2:** → “about 6.9 million visitors” (fluent but verbose).

Comparative Discussion

- **Decoder-only (GPT-2):**
 - Fluent natural text.
 - May hallucinate; weaker EM/F1.
- **Encoder-only (BERT):**
 - Best EM/F1 accuracy, efficient span extraction.
 - Cannot freely generate non-span answers.
- **Encoder-decoder (T5):**
 - Strong balance between accuracy & generation.
 - Heavier than BERT, slower inference.

Easiest to fine-tune: BERT (straightforward span prediction).

Best outputs: T5 (clean generative answers).

Most efficient: BERT (single forward pass, no decoding loop).

Part 3: Reflections

- **Real-world usage:**
 - GPT-2: story/dialogue generation.
 - BERT: extractive QA (search engines, support bots).
 - T5: summarization, open-domain QA, translation.
 - **Chain-of-Thought reasoning (conceptual):**
 - Not critical for SQuAD v1.1 (mostly factual spans).
 - Would help on multi-step reasoning QA (math, logic).
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Conclusion

Each architecture shines in different ways:

- BERT is **accurate & efficient** for extractive QA.
- T5 is **flexible & effective** for generative QA.
- GPT-2 is **fluent but less precise**.

Final takeaway: For factual extractive QA → use **BERT**; for natural generative QA → use **T5**; for free-form text → use **GPT-2**.